

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of	:	GOKARAJU et al.
	:	
For	:	ANTI-OBESE COMPOSITIONS
	:	CONTAINING HOLOPTELEA
	:	INTEGRIFOLIA EXTRACTS
	:	
Serial No.:	:	12/674,113
	:	
Filed	:	February 18, 2010
	:	
Art Unit	:	1655
	:	
Examiner	:	Melenie Lee McCormick
	:	
Att. Docket	:	LIX 3026
	:	
Confirmation No.	:	4082

AMENDMENT

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Office Action dated September 26, 2011, please amend the above-identified application as set forth below:

CLAIM AMENDMENTS begin on page 2 of this paper.

REMARKS/ARGUMENTS begin on page 10 of this paper.

CLAIM AMENDMENTS

This listing of claims will replace all prior versions and listings of claims in the application.

Listing of Claims

1. (Currently amended) Anti-adipogenic and pro-lipolytic herbal supplement ~~supplements for inhibiting or preventing or controlling adipogenesis and lipolysis involved diseases in mammals~~, comprising an extract ~~or purified fraction~~ derived from *Holoptelea integrifolia* alone or a composition comprising said extract ~~or fraction~~ optionally in combination with at least one antiobesic agent;

said extract being obtained by extraction of *Holoptelea integrifolia* leaves with water or a polar organic solvent;

said extract being present in an amount which is effective for inhibiting or controlling obesity, overweight, or hyperlipidemia in mammals..

2. (Canceled)

3. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 1, wherein said extract or purified fraction or a composition comprising said extract or fraction is used as such.

4. (Canceled)

Claims 5-15. (Canceled)

16. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 1, wherein said extract ~~or purified fraction~~ or a composition comprising said extract or fraction is formulated into a nutraceutical or pharmaceutical dosage form along with biologically acceptable excipients.

17. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 16, wherein the dosage form is a nutraceutical or pharmaceutical dosage form suitable for oral administration, and is selected from the group consisting of tablets, soft capsules, hard capsules, pills, granules, powders, emulsions, suspensions, syrups, and pellets.

18. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 16, wherein said nutraceutical or pharmaceutical dosage form is suitable for parenteral administration, and is selected from the group consisting of ~~such as~~ injections, drops, and suppositories.

19. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 16, wherein said nutraceutical or pharmaceutical dosage form is in the form of a dietary formulation selected from the group consisting of foods for specified health uses.

20. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 1, wherein the extract or purified fraction derived from *Holoptelea integrifolia* is prepared by:

- a) extracting dried *Holoptelea integrifolia* leaves ~~plant parts~~ with water or ~~an~~ a polar organic solvent or a mixture thereof to produce an extract solution;
- b) filtering the extract solution through fine filters to produce a filtrate;
- c) evaporating said filtrate to remove solvent to obtain a concentrated extract,
- d) purifying the concentrated extract to obtain said extract or purified fraction; and
- e) optionally mixing the extract or purified fraction with an anti-obesic agent or antioxidant or bio-enhancer in a blender.

21. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 20, wherein extracting dried *Holoptelea integrifolia* plant parts comprises extracting the dried *Holoptelea integrifolia* plant parts with an organic solvent selected from the group consisting of polar organic solvents selected from the group consisting of methanol, ethanol, propanol, isopropanol, n-butanol, isobutanol, tert-butanol, dimethyl ketone, methyl ethyl ketone, and methyl isobutyl ketone; ~~and non-polar organic solvents selected from~~

~~the group consisting of methyl acetate, ethyl acetate, butyl acetate, and diethyl ether.~~

22. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 1, wherein the at least one antiobesic agent is selected from the group consisting of the extracts or powders of Garcinia cambogia, green tea, green coffee bean, eucalyptus plant extract, double salt of (-)-hydroxycitric acid derived from *Garcinia* species, *Gymnema sylvestre* extract, Banaba extract, carnitine, Phaseolus vulgaris extract, bitter orange (*Citrus aurantium*) extract, Chitosan, Conjugated linoleic acid, Glucomannan (Konjac plant extract), Green coffee bean extract, Caralluma extract, Hoodia Gordonii extract, Commiphora mukul gum resin extract, Zingiber officinalis extract, Allium sativa extract, chromium (111) complexes, DHEA, and 7-Keto-DHEA.

23. (Withdrawn) A method for treating or preventing obesity or hyperlipedemia in a patient, comprising:

administering a herbal supplement according to claim 1 to said patient, wherein the herbal supplement is administered as a nutraceutical or pharmaceutical dosage form selected from the group consisting of a food, a beverage, an oral dosage form, or a parenteral dosage form.

24. (Withdrawn) A method for lowering plasma triglyceride in a patient, comprising:

administering a herbal supplement according to claim 1 to said patient, wherein the herbal supplement is administered as a nutraceutical or pharmaceutical dosage form selected from the group consisting of a food, a beverage, an oral dosage form, or a parenteral dosage form.

25. (Currently amended) The anti-adipogenic and pro-lipolytic supplement ~~supplements~~ as claimed in claim 1, further comprising an effective amount of at least one pharmaceutically or nutritionally or dietetically acceptable antioxidant, adaptogen, anti-inflammatory agent, anti-diabetic agent, bio-protectant, bioavailability enhancer, trace metal, or mixture thereof to form a formulation.

26. (Currently amended) The anti-adipogenic and pro-lipolytic supplement ~~supplements~~ as claimed in claim 1, wherein the extract or purified fraction derived from *Holoptelea integrifolia* is administered to a mammal in the form of a therapeutic formulation or food or beverage in a range from 0.01 to 300 mg/kg weight/day.

27. (Withdrawn) A method of treating obesity and, adipogenesis and lipolysis involved diseases in mammals comprising administering to a mammal in need thereof a therapeutically effective amount of a *Holoptelea integrifolia* extract or

purified fraction or a composition comprising said *Holoptelea integrifolia* extract or purified fraction, optionally in combination with at least one anti-obesic agent.

28. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 22 ~~2~~, wherein said extract or purified fraction or a composition comprising said extract or fraction is used as such.

29. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 22 ~~2~~, wherein said extract or purified fraction or a composition comprising said extract or fraction is formulated into a nutraceutical or pharmaceutical dosage form along with biologically acceptable excipients.

30. (Currently amended) The anti-adipogenic and pro-lipolytic herbal supplement ~~supplements~~ as claimed in claim 16, wherein said dosage form is a nutraceutical dosage form ~~is~~ in the form of a dietary formulation selected from the group consisting of chocolate bars; nutritional bars; creams; jams; gels; beverages selected from the group consisting of coffee, tea, milk-containing beverages, and lactic acid bacteria-containing beverages; candies, chewing gums, chocolates, gummy candies, yoghurt, ice cream, puddings, soft adzuki-bean jellies, jellies, and cookies.

31 (New) An anti-adipogenic and pro-lipolytic herbal supplement, comprising an extract derived from *Holoptelea integrifolia* alone or a composition comprising said extract optionally in combination with at least one antiobesic agent;

said extract being obtained by extraction of *Holoptelea integrifolia* leaves with water or a polar organic solvent;

said extract being present in a dosage ranging between 0.01 mg/kg weight/day and 300 mg/kg weight/day, said dosage being effective for:

- a) inhibiting adipogenesis in mammals;
- b) promoting lipolysis in mammals; or
- c) both inhibiting adipogenesis and promoting lipolysis in mammals.

32 (New) An anti-adipogenic and pro-lipolytic herbal supplement according to claim 1, wherein:

said extract derived from *Holoptelea integrifolia* is a bioenriched extract obtained by:

- a) extracting *Holoptelea integrifolia* leaves with water or a first polar organic solvent to produce a first extract;
- b) loading said first extract onto a chromatography column;
- d) washing said chromatography column with water;
- e) eluting said column with a second polar organic solvent to produce an elutant; and
- f) evaporating said polar organic solvent from the elutant to produce said bioenriched extract.

33. (New) An anti-adipogenic and pro-lipolytic herbal supplement according to claim 32, wherein said second polar organic solvent is aqueous methanol.
34. (New) An anti-adipogenic and pro-lipolytic herbal supplement according to claim 32, wherein said second polar organic solvent is 80% aqueous methanol.
35. (New) An anti-adipogenic and pro-lipolytic herbal supplement according to claim 1, wherein said effective amount ranges between 0.01 mg/kg weight/day and 300 mg/kg weight/day.
37. (New) An anti-adipogenic and pro-lipolytic herbal supplement according to claim 1, wherein said supplement contains 1% to 100% by weight of said extract derived from *Holoptelea integrifolia*.